IMAGE VIEWER HISTORY

C Programming Project using Doubly Linked List

1. Project Overview

This is a menu-driven C project that stores the names of images opened by a user. It supports opening an image, moving to the next or previous item, viewing the history, and clearing the history. A Doubly Linked List is used so each history item can point to both the previous and next item.

2. Important Note

This beginner version manages image file names and navigation history in the terminal. It does not display the actual image on screen. A graphical image viewer would need an additional graphics or image library.

3. Data Structure

Each node contains an image name, a pointer to the previous node, and a pointer to the next node. The current pointer identifies the image selected at the moment.

4. Features

• Open Image – add an image name to the history.

• Next Image – move forward through the history.

• Previous Image – move backward through the history.

• View History – display all stored image names and mark the current one.

• Clear History – delete all nodes and release memory.

• Forward-history rule – opening a new image after moving backward removes the old forward entries.

5. Complete C Program

Create a file named **image_viewer_history.c** and copy the following code.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define NAME_SIZE 100

/* Each node stores one image name and links to both neighbours. */
typedef struct Node {
    char imageName[NAME_SIZE];
    struct Node *prev;
    struct Node *next;
} Node;

Node *head = NULL;
Node *tail = NULL;
Node *current = NULL;

/* Create a new history node. */
Node *createNode(const char *name) {
    Node *newNode = (Node *)malloc(sizeof(Node));

    if (newNode == NULL) {
        printf("Memory allocation failed.\\n");
        return NULL;
    }

    strncpy(newNode->imageName, name, NAME_SIZE - 1);
    newNode->imageName[NAME_SIZE - 1] = '\\0';
    newNode->prev = NULL;
    newNode->next = NULL;

    return newNode;
}

/* Add an image to history and make it the current image. */
```

```
void openImage(void) {
    char name[NAME_SIZE];

    printf("Enter image file name (example: photo.jpg): ");
    scanf(" %99s", name);

    Node *newNode = createNode(name);
    if (newNode == NULL) {
        return;
    }

    /* Opening a new image after moving backward clears forward history. */
    if (current != NULL && current->next != NULL) {
        Node *temp = current->next;
        current->next = NULL;
        tail = current;

        while (temp != NULL) {
            Node *nextNode = temp->next;
            free(temp);
            temp = nextNode;
        }
    }

    if (head == NULL) {
        head = tail = newNode;
    } else {
        newNode->prev = tail;
        tail->next = newNode;
        tail = newNode;
    }

    current = newNode;
    printf("Opened: %s\\n", current->imageName);
}

/* Move to the next image in history. */
void nextImage(void) {
    if (current == NULL) {
        printf("No image history. Open an image first.\\n");
    } else if (current->next == NULL) {
        printf("This is the latest image.\\n");
    } else {
        current = current->next;
        printf("Current image: %s\\n", current->imageName);
    }
}

/* Move to the previous image in history. */
void previousImage(void) {
    if (current == NULL) {
        printf("No image history. Open an image first.\\n");
    } else if (current->prev == NULL) {
        printf("This is the first image.\\n");
    } else {
        current = current->prev;
        printf("Current image: %s\\n", current->imageName);
    }
}

/* Display all images from oldest to newest. */
void viewHistory(void) {
    Node *temp = head;

    if (temp == NULL) {
        printf("Image history is empty.\\n");
        return;
    }

    printf("\\n--- IMAGE VIEWING HISTORY ---\\n");
    while (temp != NULL) {
        if (temp == current) {
            printf("-> %s  [CURRENT]\\n", temp->imageName);
        } else {
            printf("   %s\\n", temp->imageName);
        }
        temp = temp->next;
    }
}

/* Delete all history nodes and release allocated memory. */
void clearHistory(void) {
    Node *temp = head;

    while (temp != NULL) {
        Node *nextNode = temp->next;
        free(temp);
        temp = nextNode;
    }

    head = tail = current = NULL;
```

```
    printf("Image history cleared.\\n");
}

int main(void) {
    int choice;

    do {
        printf("\\n===== IMAGE VIEWER HISTORY =====\\n");
        printf("1. Open Image\\n");
        printf("2. Next Image\\n");
        printf("3. Previous Image\\n");
        printf("4. View History\\n");
        printf("5. Clear History\\n");
        printf("6. Exit\\n");
        printf("Enter your choice: ");

        if (scanf("%d", &choice) != 1) {
            printf("Invalid input. Please enter a number.\\n");
            while (getchar() != '\\n') {
                /* Clear invalid input from the input buffer. */
            }
            continue;
        }

        switch (choice) {
            case 1:
                openImage();
                break;
            case 2:
                nextImage();
                break;
            case 3:
                previousImage();
                break;
            case 4:
                viewHistory();
                break;
            case 5:
                clearHistory();
                break;
            case 6:
                clearHistory();
                printf("Thank you for using Image Viewer History!\\n");
                break;
            default:
                printf("Invalid choice. Try again.\\n");
        }
    } while (choice != 6);

    return 0;
}
```

6. How to Compile and Run

Using GCC in the VS Code terminal or Command Prompt:

```
gcc image_viewer_history.c -o image_viewer_history
image_viewer_history
```

On Windows, run the program with:

```
image_viewer_history.exe
```

7. Sample Output

```
===== IMAGE VIEWER HISTORY =====
1. Open Image
2. Next Image
3. Previous Image
4. View History
5. Clear History
6. Exit
Enter your choice: 1
Enter image file name: photo1.jpg
Opened: photo1.jpg

Enter your choice: 1
Enter image file name: holiday.png
Opened: holiday.png

Enter your choice: 4

--- IMAGE VIEWING HISTORY ---
   photo1.jpg
-> holiday.png  [CURRENT]
```

8. Conclusion

The Image Viewer History project demonstrates how a Doubly Linked List can be used to maintain browsing history. It provides practice with structures, pointers, dynamic memory allocation, insertion, traversal, deletion, and freeing memory in C.